Automated ETL & Dashboard Analysis Report

Dataset: Raw_Data.csv
Domain: Finance
Generated: 06 November 2026, 16:23

1. Dataset Overview

Metric	Value
Rows	4,535
Columns	10
Rows after cleaning	4,535
Missing cells after cleaning	0
Duplicate rows after cleaning	2

2. ETL Process

• Raw data uploaded and inspected.
• Column names, data types, missing values, and duplicates reviewed.
• Cleaning actions applied only after user selection.
• Data quality checks performed on the working dataset.
• Exploratory analysis and visualizations generated.
• Interactive dashboard prepared with available filters and KPIs.
No manual cleaning actions were recorded.

3. Data Quality

Column	Type	Missing	Unique
Title	str	0	4500
Budget	int64	0	491
ID	int64	0	4527
Original Language	str	0	42
Popularity	float64	0	4531
Release Date	str	0	2613
Revenue	int64	0	4379
Runtime	float64	0	136
Vote Average	float64	0	70
Vote Count	int64	0	1585

4. KPI Summary

KPI	Value
Total Budget	■129.97 B
Average Budget	■28.66 M
Average Vote Average	6.19
Analysis Period	2000-01 – 2017-12

5. Key Findings

1. The working dataset contains 4,535 records across 10 columns.
2. No missing cells were detected in the current working dataset.
3. 2 fully duplicated rows (0.04% of records) were found.
4. Identifier candidate 'ID' has 16 rows with repeated values and 0 missing values; uniqueness is not confirmed.
5. 'Budget' ranges from 0.00 to 380,000,000.00, with mean 28,659,347.60 and median 12,000,000.00.
6. 'Budget' is positively skewed (skewness 2.47); a small number of high values may influence the mean.
7. 'Release Date' spans 2000-01-02 through 2017-12-05 (2,613 distinct date values parsed).

6. Recommendations

1. Inspect duplicate records and confirm whether they are repeated events or accidental copies before removal.
2. Validate whether 'ID' should uniquely identify rows; if not, define a composite key or document valid repeats.
3. Compare median and mean for 'Budget' and review the highest-value records before making typical-value claims.
4. Use 'Release Date' for trend analysis after confirming it represents the relevant business event date and checking for gaps.

7. Visual Analysis

Chart: Revenue over time (image export unavailable; chart data remains summarized above).
Chart: Distribution of Revenue (image export unavailable; chart data remains summarized above).
Chart: Numeric correlation (image export unavailable; chart data remains summarized above).

8. Conclusion

The dataset was processed through a structured analyst workflow covering data inspection, cleaning, quality validation, analysis, visualization, and reporting. Final business decisions should be based on validated definitions and stakeholder context.